

The two polynomial capacities really differ

Candidate counterexample packet for arXiv:math/0401078

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Outcome. For every $m \geq 2$ and $0 \leq k \leq m - 2$, there are one-dimensional vector subspaces $\mathcal{A} \subset W^{m,p}(Q)$ for which

$$\Theta_{m,k,p}^\alpha(\mathcal{A}) = 0 \quad \text{but} \quad \Gamma_{m,k,p}(\mathcal{A}) > 0.$$

The positive Γ value may be made arbitrarily small. This supplies the instances requested in the source and shows that no uniform two-sided equivalence can hold below the already-known case $k = m - 1$.

1. The question and the definitions

For a cube Q , let $\mathcal{P}_k$ be the polynomials of degree at most k . In the notation of the source,

$$U_{P,\mathcal{A}} = \{w \in W_0^{m,p}(2Q) : w|_Q - P \in \mathcal{A}\}$$

and Maz'ya's polynomial capacity is

$$\Gamma_{m,k,p}(\mathcal{A}) = \inf_{0 \neq P \in \mathcal{P}_k} \inf_{w \in U_{P,\mathcal{A}}} \frac{\|\nabla^m w\|_{L^p(2Q)}^p}{\|P\|_{L^p(Q)}^p}. \quad (1)$$

The source's Θ^α first restricts to those $u \in \mathcal{A}$ for which the degree- $\leq k$ part of the polynomial projection dominates its higher-degree complement by the factor α , and whose nonpolynomial remainder is at most half the polynomial part. It then takes

$$\Theta_{m,k,p}^\alpha(\mathcal{A}) = \min \left\{ 1, \inf_u \frac{\|\nabla^m u\|_{L^p(Q)}^p}{\|\Pi_{m-1,k,p} u\|_{L^p(Q)}^p} \right\}. \quad (2)$$

Now we turn back to a discussion of the case in the general setting.

The technicalities in the definition of the polynomial capacity Θ that takes the place of the Maz'ya polynomial capacity Γ are more difficult. In spite of this many tries for good estimates have turned out rather well. This after harder work i.e. One can say that as long as the polynomial capacity Θ has the same estimate as the polynomial capacity Γ , the second Poincaré inequality in fact provides a stronger result. This is of course obvious.

What is not obvious that we do not know instances where these polynomial capacities really are different. Hence we here have an open question.

The situation that the set $\mathcal{A}$ is not a vector space turns up in the treatment of Θ for $\mathcal{A}$ a nonnegative cone. The implication is that nonnegative functions sometimes have smaller (= better) constant C in the inequality than some related set of general functions. This key result will be used for the last part of this paper as well as be very important in the coming study of Hardy inequalities.

Figure 1: The exact open question on source PDF page 6.

2. A separation for every nontrivial index

Theorem 1. Let Q be a cube, $1 \leq p < \infty$, $m \geq 2$, $0 \leq k \leq m - 2$, and $\alpha > 0$. There are one-dimensional vector subspaces $\mathcal{A}_\varepsilon$ such that

$$\Theta_{m,k,p}^\alpha(\mathcal{A}_\varepsilon) = 0 < \Gamma_{m,k,p}(\mathcal{A}_\varepsilon) < \infty.$$

Moreover $\Gamma_{m,k,p}(\mathcal{A}_\varepsilon) \leq C\varepsilon^p$, so the positive capacity can tend to zero.

Proof. Choose $0 \neq P_0 \in \mathcal{P}_k$ and a polynomial $H \in \mathcal{P}_{m-1} \setminus \mathcal{P}_k$ having no terms of degree at most k ; a homogeneous polynomial of degree $k+1$ will do. Put

$$a_\varepsilon = P_0 + \varepsilon H, \quad \mathcal{A}_\varepsilon = \text{span}\{a_\varepsilon\},$$

where $\varepsilon > 0$ is small enough that $\|P_0\|_p \geq \alpha \varepsilon \|H\|_p$.

For every nonzero t , the exact polynomial ta_ε has projection itself, degree- $\leq k$ part tP_0 , complement $t\varepsilon H$, and zero nonpolynomial remainder. It therefore belongs to the restricted test set in (2). Since $\deg a_\varepsilon \leq m-1$, $\nabla^m a_\varepsilon = 0$, and hence

$$\Theta_{m,k,p}^\alpha(\mathcal{A}_\varepsilon) = 0.$$

We prove that $\Gamma > 0$. The higher-order Poincaré inequality on $2Q$ gives

$$\|\nabla^m w\|_{L^p(2Q)} \geq c_Q \|w\|_{L^p(Q)} \quad (w \in W_0^{m,p}(2Q)). \quad (3)$$

Define the finite-dimensional angle

$$\delta_\varepsilon = \inf_{\substack{P \in \mathcal{P}_k, \\ t \in \mathbb{R}, \|P\|_p=1}} \|P + t(P_0 + \varepsilon H)\|_{L^p(Q)}.$$

This number is positive. Indeed, if a minimizing sequence tended to zero, passing to the quotient $\mathcal{P}_{m-1}/\mathcal{P}_k$ would first force $t\varepsilon[H] \rightarrow 0$, hence $t \rightarrow 0$; the original relation would then force $P \rightarrow 0$, contradicting $\|P\|_p = 1$.

For $w \in U_{P,\mathcal{A}_\varepsilon}$ we have $w|_Q = P + ta_\varepsilon$ for some t . Scaling P and using (3) therefore shows

$$\frac{\|\nabla^m w\|_p^p}{\|P\|_p^p} \geq c_Q^p \delta_\varepsilon^p > 0.$$

Taking the two infima in (1) proves positivity.

Finally, let $\eta \in C_c^\infty(2Q)$ equal one on Q . Take $P = -P_0$ and $w = \varepsilon \eta H$. Then $w|_Q - P = a_\varepsilon \in \mathcal{A}_\varepsilon$, so this is admissible and gives

$$\Gamma_{m,k,p}(\mathcal{A}_\varepsilon) \leq \varepsilon^p \frac{\|\nabla^m(\eta H)\|_{L^p(2Q)}^p}{\|P_0\|_{L^p(Q)}^p}.$$

This also proves finiteness and the asserted small-capacity estimate. $\square$

3. The smallest explicit example

Explicit example 1. Take

$$Q = (-1, 1), \quad m = 2, \quad k = 0, \quad p = 2, \quad \alpha = 1, \quad \mathcal{A} = \text{span}\{1 + x/2\}.$$

The constant and linear parts have squared norms 2 and $1/6$, so the Θ dominance test holds, and $\Theta_{2,0,2}^1(\mathcal{A}) = 0$.

For the Γ lower bound, write a constant polynomial as $P = c$ and the line parameter as t . After dividing by $c \neq 0$, the normalized squared trace norm is

$$\frac{\|1 + t(1 + x/2)\|_{L^2(-1,1)}^2}{\|1\|_{L^2(-1,1)}^2} = (1 + t)^2 + \frac{t^2}{12}.$$

Its minimum is $1/13$. Thus (3) gives the explicit strict bound

$$\Gamma_{2,0,2}(\mathcal{A}) \geq c_Q^2/13 > 0.$$

4. Boundary and novelty

The construction works throughout $k \leq m-2$. At $k = m-1$ there is no higher polynomial component available, and the source already proves $\Theta \asymp \Gamma$; the example therefore separates exactly the remaining index range. It uses a vector subspace, not an artificial non-vector set.

The result answers the capacities question as defined for arbitrary subsets of $W^{m,p}(Q)$. It does not decide whether the capacities might be equivalent on narrower geometric families such as restrictions of $W_0^{m,p}(K^c)$ used in spectral synthesis.

Bounded primary-source searches through 17 August 2026 found the source and its 2004 follow-up. The latter repeats the equivalence question; no later source recording this separation was found. External expert review is still recommended before dissemination.

References

- [1] A. Wannebo, *Polynomial capacities, Poincaré type inequalities and Spectral synthesis in Sobolev space*, arXiv:math/0401078 (2004).
- [2] A. Wannebo, *Hardy and Hardy PDO type inequalities in domains. Part I*, arXiv:math/0401253 (2004).